

Algebra II with Trigonometry

Chapter 6.1

Mental Math (Gemini)

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Direction of Rotation 1. If you are marking off a distance t on the unit circle and t is a negative number, in which direction do you travel from the starting point?

Basic Circumference 2. What is the total distance traveled if you move exactly one quarter of the way around the unit circle?

Axis Terminal Point 3. What are the (x, y) coordinates of the terminal point determined by $t = \frac{3\pi}{2}$?

Identifying Quadrants 4. If a terminal point on the unit circle has a negative x-coordinate and lies below the x-axis, in which quadrant does it lie?

Missing Coordinate Calculation 5. A terminal point P lies in Quadrant I on the unit circle. If its x-coordinate is $\frac{3}{5}$, what is its y-coordinate?

Reference Number Axis 6. When finding the reference number for a terminal point located in Quadrant II, do you measure the distance to the positive x-axis or the negative x-axis?

Calculating Reference Numbers 7. What is the reference number for the real number $t = 9\pi$?

Coterminal Points 8. If the terminal point for a real number t is (a, b) , what is the terminal point for $t + 2\pi$?

Terminal Point Symmetry 9. Suppose the terminal point determined by t is $(\frac{3}{5}, \frac{4}{5})$. What is the terminal point determined by $-t$?

Unit Circle Constraints 10. If a point on the unit circle has an x-coordinate of 1, what must its y-coordinate be?